GEOMETRY NOTES

Triangle basics, medians and centres

A median joins a vertex to the midpoint of the opposite side. If D is the midpoint of BC, then BD = DC and AD is a median.

$$AG : GD = 2 : 1$$

The three medians meet at the centroid G. The centroid is two-thirds of the way from each vertex along its median.

$$G = ((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$$

Incentre: intersection of internal angle bisectors. Circumcentre: intersection of perpendicular bisectors. Orthocentre: intersection of altitudes.

$$\angle A + \angle B + \angle C = 180°$$

The perpendicular bisector of a side passes through its midpoint and is perpendicular to that side.

In an isosceles triangle, equal sides have equal opposite angles. The line from the apex to the midpoint of the base is also an altitude and angle bisector.

In an equilateral triangle, all sides are equal and every angle is 60°.

$$\text{Equilateral triangle: } h = (\sqrt{3}/2)a; \text{ Area} = (\sqrt{3}/4)a^2$$

Triangle exterior angle equals the sum of the two opposite interior angles.

$$\angle ACD \text{ (exterior)} = \angle A + \angle B$$

Ceva’s theorem

In triangle ABC, D lies on BC, E on CA and F on AB. AD, BE and CF are concurrent if and only if:

$$(BD/DC) \times (CE/EA) \times (AF/FB) = 1$$

Angle bisector theorem

If AD bisects ∠A and D lies on BC:

$$BD/DC = AB/AC$$

$$AD^2 = AB \times AC - BD \times DC$$

Cosine rule:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\cos A = (AB^2 + AC^2 - BC^2)/(2 \times AB \times AC)$$

Triangle area

$$\text{Area} = ½ \times \text{base} \times \text{perpendicular height}$$

For side lengths a, b and c, semi-perimeter s is:

$$s = (a + b + c)/2$$

$$\text{Heron’s formula: } \Delta = \sqrt{[s(s - a)(s - b)(s - c)]}$$

$$\Delta = ½ab \sin C = ½bc \sin A = ½ca \sin B$$

For a right triangle with perpendicular sides a and b:

$$\Delta = ½ab$$

For a triangle with inradius r and circumradius R:

$$\Delta = rs = abc/(4R)$$

Here Δ is the area and s is the semi-perimeter.

Pythagoras and right-triangle results

$$c^2 = a^2 + b^2$$

Common right-triangle triples: 3-4-5; 5-12-13; 8-15-17; 7-24-25.

For a right triangle, the median to the hypotenuse equals half the hypotenuse.

$$\text{Median to hypotenuse} = c/2$$

If the altitude from the right angle divides the hypotenuse into segments p and q:

$$h^2 = pq; \; a^2 = c \times p; \; b^2 = c \times q$$

Similarity and proportionality

Similar triangles have equal corresponding angles and proportional corresponding sides.

$$AB/DE = BC/EF = AC/DF$$

$$\text{Area ratio} = (\text{corresponding side ratio})^2$$

If DE ∥ BC in triangle ABC, with D on AB and E on AC:

$$AD/DB = AE/EC;\ AD/AB = AE/AC = DE/BC$$

Corresponding altitudes, medians and angle-bisector lengths of similar triangles are in the same ratio as corresponding sides.

For similar figures, the ratio of perimeters equals the side ratio; the ratio of areas equals the square of the side ratio.

Parallel lines and intercept theorem

When parallel lines cut two transversals, corresponding intercepted segments are proportional.

$$a/b = c/d$$

Match the segments between the same pair of parallel lines.

Ladder and cross-ladder

For the standard cross-ladder arrangement between vertical walls and a level floor, if ladders touch the walls at heights a and b and cross at height h:

$$1/h = 1/a + 1/b \Rightarrow h = ab/(a + b)$$

Triangle and quadrilateral formulae

Quadrilaterals

Parallelogram: opposite sides are parallel and equal; opposite angles are equal; adjacent angles sum to 180°; diagonals bisect each other.

$$\text{Area} = \text{base} \times \text{height} = ab \sin\theta$$

Rectangle: all angles are 90°; opposite sides are equal; diagonals are equal and bisect each other.

$$\text{Area} = lb;\ \text{perimeter} = 2(l + b);\ \text{diagonal} = \sqrt{(l^2 + b^2)}$$

Square: all sides are equal; all angles are 90°; diagonals are equal, perpendicular and bisect the vertex angles.

$$\text{Area} = a^2;\ \text{perimeter} = 4a;\ \text{diagonal} = a\sqrt{2}$$

Rhombus: all sides are equal; diagonals bisect each other at right angles.

$$\text{Area} = \tfrac{1}{2}d_1d_2 = \text{base} \times \text{height}$$

Trapezium: parallel sides are a and b, height is h.

$$\text{Area} = \tfrac{1}{2}(a + b)h$$

For a trapezium, the segment joining the midpoints of the non-parallel sides has length:

$$\text{Mid-segment} = (a + b)/2$$

Kite: two pairs of adjacent sides are equal; diagonals are perpendicular.

$$\text{Area of kite} = \tfrac{1}{2}d_1d_2$$

Circle geometry

$$\text{Circumference} = 2\pi r = \pi d$$

$$\text{Area} = \pi r^2$$

A radius to the point of tangency is perpendicular to the tangent. Tangents drawn from the same external point are equal.

$$PA = PB \text{ (tangents from external point P)}$$

A perpendicular from the centre to a chord bisects the chord. Equal chords are equidistant from the centre and subtend equal angles at the centre.

The angle subtended by a diameter at the circumference is 90°.

Angles subtended by the same chord in the same segment are equal.

The angle at the centre is twice the angle at the circumference subtended by the same arc.

$$\text{Central angle} = 2 \times \text{inscribed angle}$$

Opposite angles of a cyclic quadrilateral are supplementary.

$$\angle A + \angle C = 180°;\ \angle B + \angle D = 180°$$

An exterior angle of a cyclic quadrilateral equals the interior opposite angle.

$$\text{Ptolemy: } AC \times BD = AB \times CD + AD \times BC$$

Intersecting chords inside a circle:

$$PA \times PB = PC \times PD$$

Two secants from an external point:

$$\text{External segment} \times \text{whole secant} = \text{external segment} \times \text{whole other secant}$$

Tangent-secant theorem:

$$\text{Tangent length}^2 = \text{external secant segment} \times \text{whole secant}$$

For a chord of length c at perpendicular distance d from a circle's centre, with radius r:

$$(c/2)^2 + d^2 = r^2$$

Polygons

$$\text{Interior-angle sum} = (n - 2) \times 180°$$

$$\text{Each interior angle of a regular n-gon} = [(n - 2) \times 180°]/n$$

$$\text{Sum of exterior angles} = 360°$$

$$\text{Each exterior angle of a regular n-gon} = 360°/n$$

$$\text{Number of diagonals} = n(n-3)/2$$

Examples: regular pentagon exterior angle = 72°; hexagon = 60°; heptagon = 360°/7; octagon = 45°.

Regular octagon with side length a

$$\text{Area} = 2a^2(1+\sqrt{2})$$

$$\text{Small diagonal} = a\sqrt{(2+\sqrt{2})}$$

$$\text{Long diagonal} = \sqrt{2} \times \text{small diagonal}$$

$$\text{Middle diagonal} = a(1+\sqrt{2})$$

Regular polygon area

$$\text{Area} = \tfrac{1}{2} \times \text{perimeter} \times \text{apothem}$$

For a regular n-gon of side a, perimeter = na. The apothem is the perpendicular distance from the centre to a side.

$$\text{Area} = na \times \text{apothem} / 2$$

Coordinate geometry and additional triangle results

Coordinate geometry

Distance between (x_1,y_1) and (x_2,y_2): $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$

Slope = $(y_2 - y_1)/(x_2 - x_1)$, when $x_2 \neq x_1$

Line through (x_1,y_1): $y - y_1 = m(x - x_1)$

Centroid of triangle = $((x_1+x_2+x_3)/3, (y_1+y_2+y_3)/3)$

If P divides AB internally in ratio AP:PB = m:n:

$P = ((n\,x_1 + m\,x_2)/(m+n), (n\,y_1 + m\,y_2)/(m+n))$

Area of triangle with vertices (x_1,y_1), (x_2,y_2), (x_3,y_3):

$\Delta = \frac{1}{2}|x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)|$

Additional triangle identities

Triangle inequality:

$|b - c| < a < b + c$

Apollonius' theorem, where AD is a median and D is the midpoint of BC:

$AB^2 + AC^2 = 2(AD^2 + BD^2)$

Inradius and circumradius:

$r = \Delta/s$; $R = abc/(4\Delta)$

For any triangle, the longest side is opposite the largest angle.

For a triangle with sides a, b, c opposite angles A, B, C:

Sine rule: $a/\sin A = b/\sin B = c/\sin C = 2R$

Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$

The median from A to side BC:

$AD^2 = (2AB^2 + 2AC^2 - BC^2)/4$

Mensuration: two-dimensional shapes

Rectangle: area = lb; perimeter = 2(l+b)

Square: area = a^2; perimeter = 4a

Triangle: area = $\frac{1}{2}bh$

Parallelogram: area = bh

Trapezium: area = $\frac{1}{2}(a+b)h$

Circle: area = πr^2; circumference = $2\pi r$

Sector area = $(\theta/360°)\pi r^2$

Arc length = $(\theta/360°)2\pi r$ (θ in degrees)

Annulus area = $\pi(R^2 - r^2)$

Mensuration: three-dimensional shapes

Cube: volume = a^3; total surface area = $6a^2$; lateral surface area = $4a^2$

Cuboid: volume = lbh; total surface area = 2(lb + bh + hl)

Cylinder: volume = $\pi r^2 h$; curved area = $2\pi rh$; total area = $2\pi r(h+r)$

Cone: volume = $\frac{1}{3}\pi r^2 h$; slant height $\ell = \sqrt{(r^2+h^2)}$; curved area = $\pi r\ell$

Sphere: volume = $\frac{4}{3}\pi r^3$; surface area = $4\pi r^2$

Hemisphere: volume = $\frac{2}{3}\pi r^3$; curved area = $2\pi r^2$; total area = $3\pi r^2$

Geometry topic from the original notes: star polygons

The star-polygon expression on the handwritten page is too unclear to reproduce confidently. It is left unspecified rather than guessed.

Formula summary

Triangle angle sum = 180°; area = $\frac{1}{2}bh$

Heron: $\Delta = \sqrt{[s(s-a)(s-b)(s-c)]}$, $s = (a+b+c)/2$

Pythagoras: $c^2 = a^2+b^2$

Ceva: $(BD/DC)(CE/EA)(AF/FB) = 1$

Angle bisector: $BD/DC = AB/AC$

Circle: area $= \pi r^2$; circumference $= 2\pi r$

Cyclic quadrilateral: opposite angles sum to $180°$

Ptolemy: $AC \times BD = AB \times CD + AD \times BC$

Polygon interior sum $= (n-2)180°$; diagonals $= n(n-3)/2$

Similar-triangle area ratio = (side ratio)^2